RANGKUMAN BAB 3

NAMA : Bagea Alifa

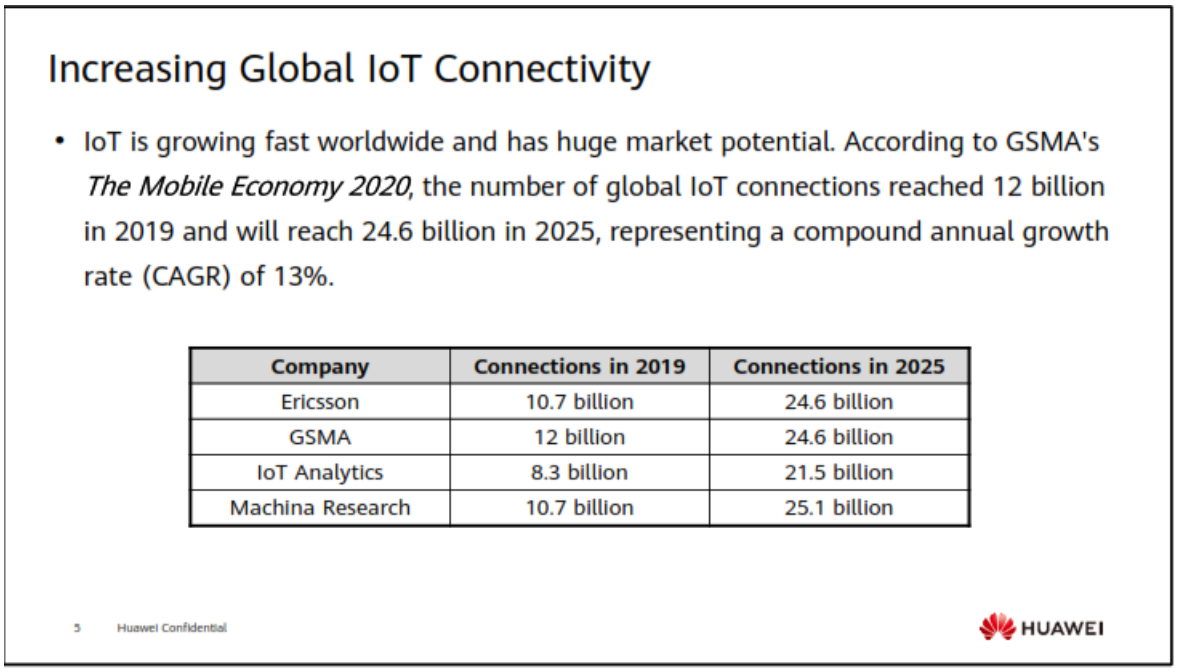
: Bahrani

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IoT Today

1. IoT industry trends

- Global IoT connections reached 12 billion in 2019 and are projected to reach 24.6 billion by 2025.



Increasing Global IoT Connectivity

- IoT is growing fast worldwide and has huge market potential. According to GSMA's *The Mobile Economy 2020*, the number of global IoT connections reached 12 billion in 2019 and will reach 24.6 billion in 2025, representing a compound annual growth rate (CAGR) of 13%.

Company	Connections in 2019	Connections in 2025
Ericsson	10.7 billion	24.6 billion
GSMA	12 billion	24.6 billion
IoT Analytics	8.3 billion	21.5 billion
Machina Research	10.7 billion	25.1 billion

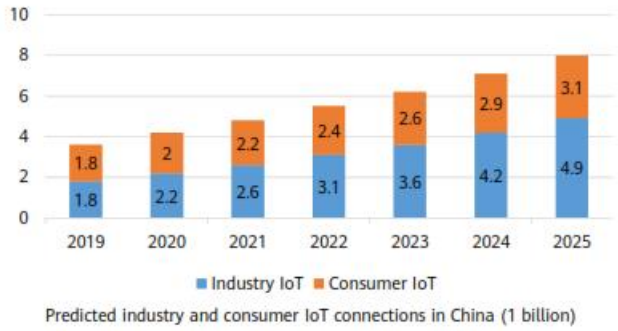
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- The share of industry IoT connections is increasing compared to consumer IoT and is expected to account for 61.2% of total connections by 2025. Key growth areas are smart manufacturing, transportation, health, and energy.

Predicted Industry and Consumer IoT Connections in China

- Industry IoT connections will take up more of the total IoT connections because more and more industries start using IoT for digitalization. According to GSMA Intelligence, the number of industry IoT devices will exceed that of consumer IoT devices by 2024.



Predicted industry and consumer IoT connections in China (1 billion)

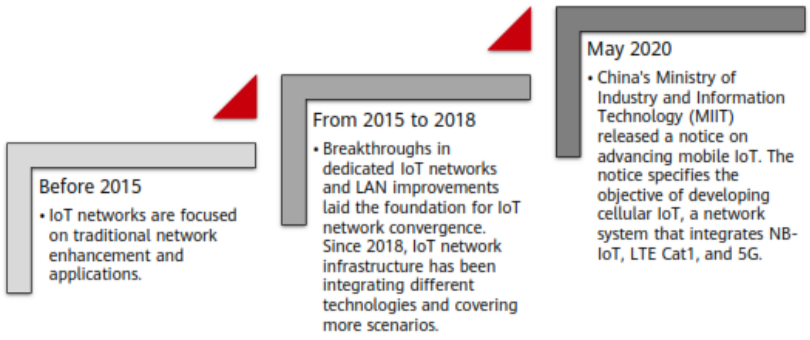
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2. IoT infrastructure integration

- Cellular IoT (NB-IoT, LTE Cat1, 5G) is leading network integration efforts.

Evolution into Space-Air-Ground Integrated Networks

- Mobile networks (cellular IoT and unlicensed IoT networks), local area networks (LANs), satellite networks, drones, and hot air balloons form global IoT networks that integrate space, air, and ground. They provide reliable access for global IoT applications anytime, anywhere.



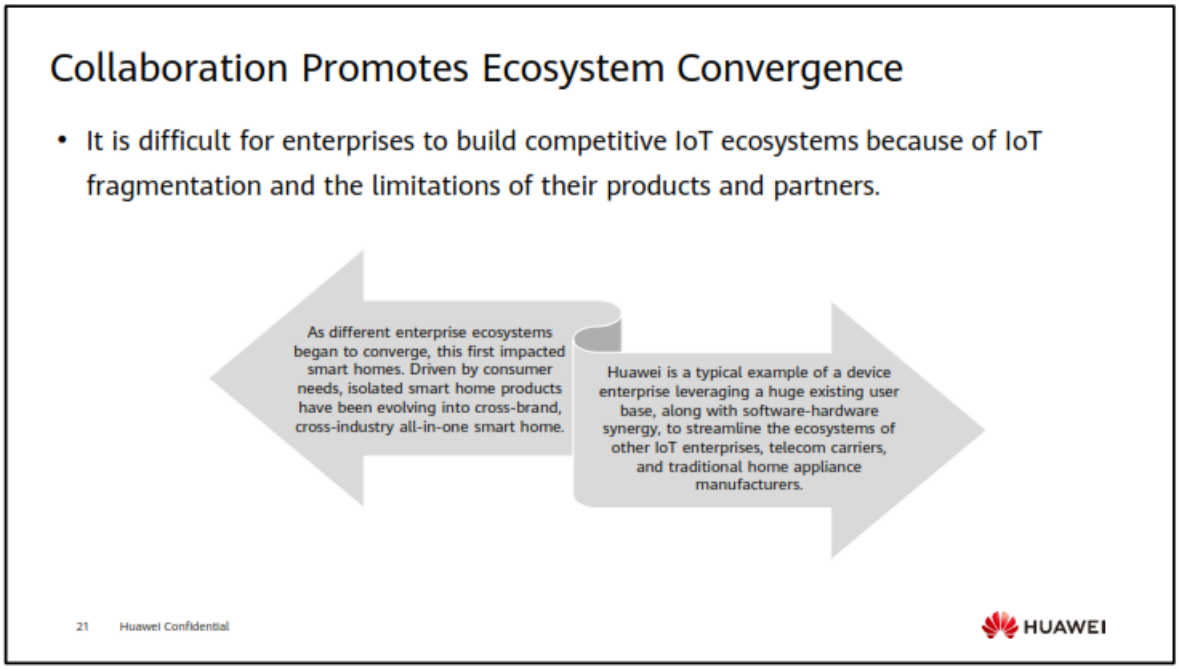
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- Integration across IoT infrastructure layers (devices, platforms, networks) and external ecosystems is accelerating.

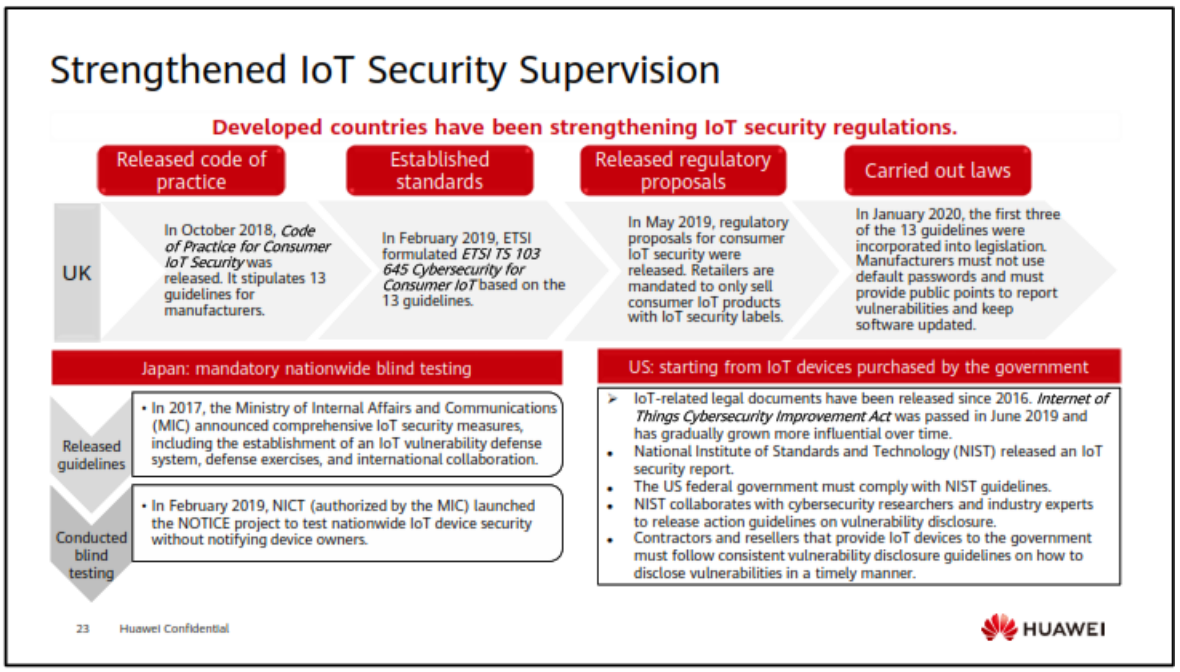
- Customizable infrastructure and intent-driven networks that adapt to usage scenarios are emerging.

3. IoT ecosystem expansion and security

- Enterprises are collaborating to build interoperable ecosystems, leverage open source, and standardize data models/profiles.

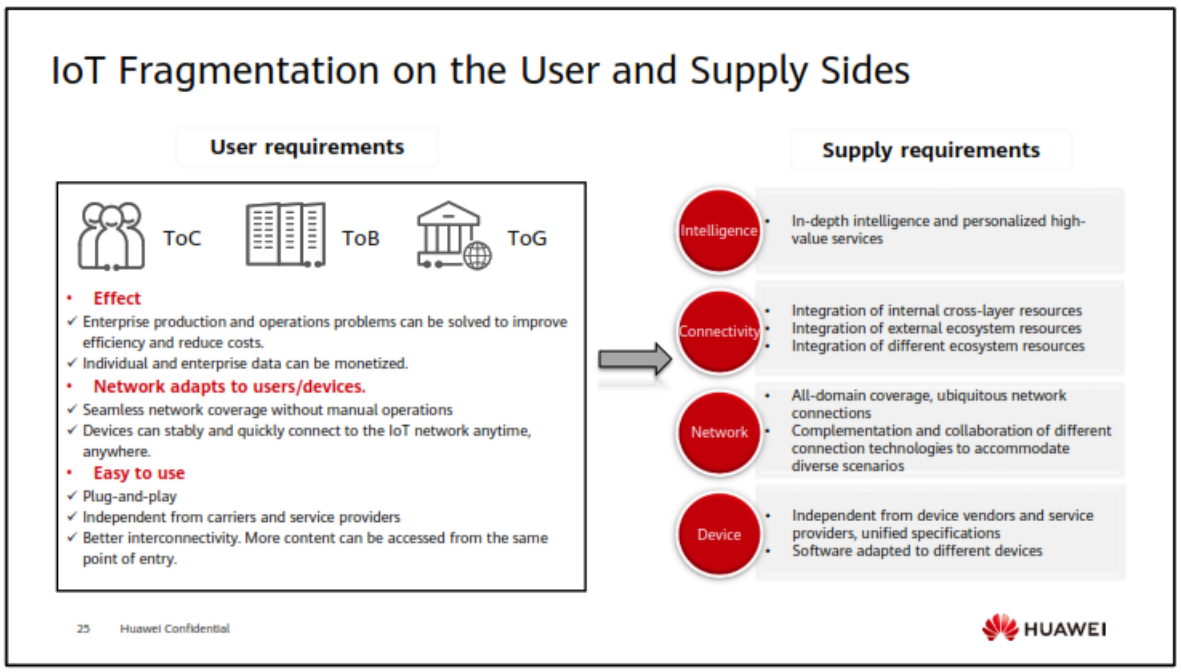


- Countries have introduced security regulations and IoT product certification programs. Blind product security testing is also being conducted.



4. IoT fragmentation issues

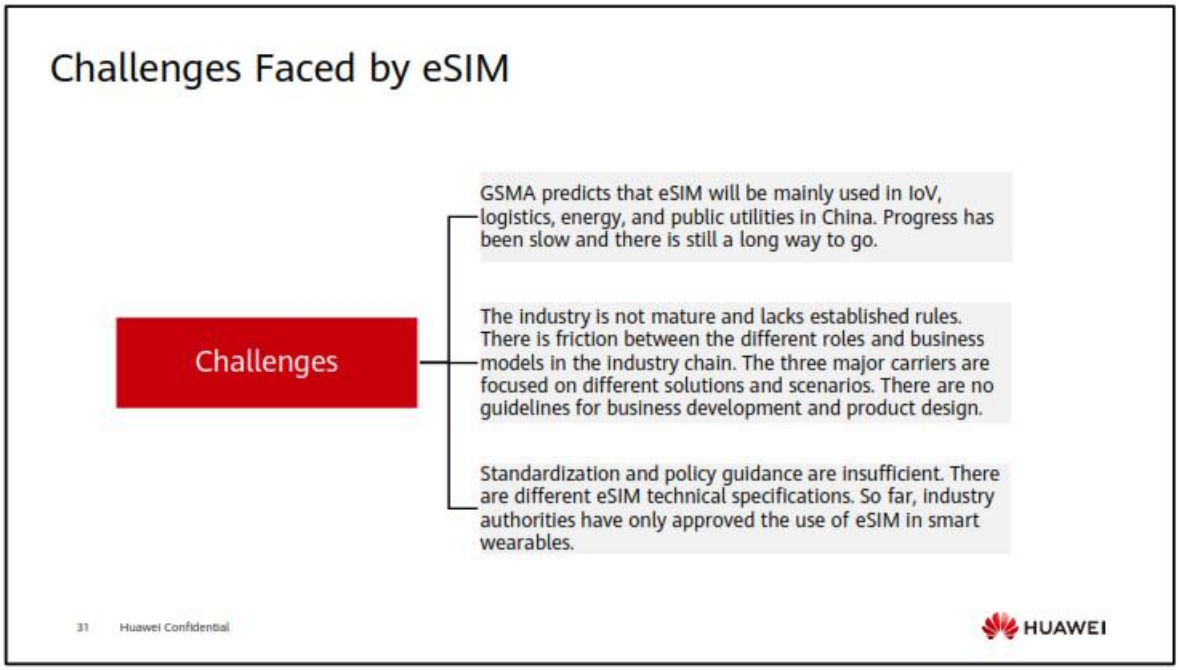
- Fragmented requirements across consumer/enterprise/government sectors and complex, heterogeneous technologies.



- Frequent security issues and increasing risks. High costs and unclear business models.

5. Potential solutions

- eSIM allows flexible carrier switching and enriches IoT products and services.



- IPv6 provides the scalable addressing and mobility needed for massive IoT growth.

- Integrated network infrastructure with space/air/ground coverage to adapt to usage scenarios.

- AI + IoT (AIoT) enables intelligent analysis and applications.